

# Evaluating Learning Trajectories of Neural Morphology Acquisition Models

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## Abstract

Computational models of morphology acquisition have played a central role in debates over the nature of morphological representations since the origin of the Past Tense Debate in the 1980s. The apparent success of recent artificial neural network architectures for morphological inflection in natural language processing has revitalized this debate. However, despite their often good performance, the actual suitability of these advanced neural networks as models of human morphology acquisition remains uncertain. We argue that much of this confusion stems from inconsistent methods of training and evaluation. In this work, we demonstrate that more careful data set creation and an evaluation combining quantitative analysis and comparison with human development puts the evaluation of neural models on firmer ground.

## 1 Introduction

Computational models of morphological inflection burst onto the scene as part of the Past Tense Debate in the late 1980s (Rumelhart and McClelland, 1986; Pinker and Prince, 1988), where they were seen as providing insights into the cognitive computations underlying morphological learning and representation (see Pinker and Ullman (2002); McClelland and Patterson (2002); Seidenberg and Plaut (2014) for surveys). One feature of the debate was a push on one side for connectionist models of inflection: artificial neural networks (ANNs) that explicitly aimed to model human cognition. The Past Tense Debate subsided over the years, but with the advent of more powerful “deep” ANNs in machine learning and natural language processing (NLP) over the last decade, there has been renewed interest in ANNs as potential cognitive models of morphology learning and representation (Kirov and Cotterell, 2018; Corkery et al., 2019; McCurdy et al., 2020; Belth et al., 2021; Beser, 2021; Dankers et al., 2021; Wiemerslage et al., 2022; Kodner et al., 2023a).

At the same time, morphological inflection has established itself as a (relatively) standardized task in NLP in the guise of the SIGMORPHON shared tasks (Cotterell et al., 2016, 2017, 2018; McCarthy et al., 2019; Vylomova et al., 2020; Pimentel et al., 2021; Kodner et al., 2022; Goldman et al., 2023).<sup>1</sup> While submissions to these competitions are engineered primarily to maximize accuracy (and nearly saturate performance on many data sets) rather than contribute to a scientific understanding of cognition, the 2021, 2022, and 2023 competitions included explicitly cognitive sub-tasks (Kodner and Khalifa, 2022; Breiss and Jo, 2023).<sup>2</sup> Though the details of the task setup have evolved over time, the modern standard task is fundamentally similar to the one of decades past: given a set of lemmas and inflected forms, learn how to predict inflected forms.

The literature on the cognitive applicability of recent ANNs remains contradictory (e.g., Kirov and Cotterell (2018), but see Corkery et al. (2019)). We suspect that this is partly explained by inconsistent evaluation criteria. In terms of accuracy, performance on the shared tasks is often near-ceiling. However, when ANNs are instead evaluated by comparing their ratings of inflected forms to human acceptability

<sup>1</sup>The 2024 shared task overview paper is unpublished <https://github.com/sigmorphon/2024InflectionST>

<sup>2</sup>The 2021 competition did not yield a separate publication for these tasks <https://github.com/sigmorphon/2021Task0>

judgments, correlations are problematic for specific phenomena close to the original Past Tense Debate (Clahsen and Rothweiler, 1993), including English past tense (Corkery et al., 2019) and German noun pluralization (McCurdy et al., 2020).

Another alternative to pure accuracy or correlation with acceptability judgments is to evaluate the ‘human-likeness’ of productions. This can be evaluated by analyzing learning trajectories and categorizing errors. Such evaluation has a long history—it was part of the evidence that Rumelhart and McClelland (1986) marshaled in favor of their connectionist model at the start of the debates and was integral to Pinker and Prince’s (1988) early counterargument as well. Taking this approach, however, even state-of-the-art ANNs submitted to the shared task come up lacking. For example, no evidence for the well-known *u*-shaped learning of English past tense (Marcus et al., 1992; Prasada and Pinker, 1993) was observed by Belth et al. (2021) or Kodner and Khalifa (2022).<sup>3</sup>

## 2 Research Goals

We aim to evaluate ANNs in terms of their overall accuracy and match to human learning trajectories. This approach takes inspiration from the developmental literature and early papers in the Past Tense Debate where errors were taken to provide particular insight into the representations underlying learning. We investigate the acquisition of English past tense (Marcus et al., 1992), German noun pluralization (Marcus et al., 1995), and Arabic noun pluralization (Ravid and Farah, 1999), since much is known about acquisition patterns for these phenomena.

Our main methodological contribution is the creation of more developmentally-plausible training data that can slot directly into the now-standard task setup. Training data is sampled from child-directed speech from CHILDES (MacWhinney, 2000) to the extent possible, following Belth et al. (2021) and Kodner and Khalifa (2022). Most SIGMORPHON shared tasks have sampled data from the UniMorph database of inflectional patterns, which is extracted primarily from Wiktionary (McCarthy et al., 2020; Batsuren et al., 2022) or from the CELEX corpus (Baayen et al., 1996), such as Kirov and Cotterell (2018) and Wiemerslage et al. (2022). Neither data source corresponds well to early learner input.

Estimates of child vocabulary knowledge for many languages (Fenson et al., 1994; Bornstein et al., 2004; Szagun et al., 2006), combined with observations of the timeline of morphological development (e.g., Brown, 1973; Aksu-Koç, 1985; Ravid and Farah, 1999; Elsen, 2002; Deen, 2005) show that children acquire most of their morphological competence on the basis of just hundreds of types regardless of a language’s morphological complexity.<sup>4</sup> Thus, we limit our training data to at most 1000 lemmas. While incremental learning would most closely reflect language acquisition, current models favor batch approaches. To somewhat approximate incremental learning, we train models on a series of nested training sets of increasing size.

Additionally, we compare frequency-weighted and uniform train-test splitting. Frequency-weighted splitting more closely approximates the human learning task and real-world applications, since vocabulary acquisition is correlated with item frequency in the input (Goodman et al., 2008; Swingley and Humphrey, 2018; Braginsky et al., 2019). Children need to use their productive morphological knowledge to inflect forms that they have not already seen in their input, and these tend to be low-frequency forms. However, most cognitive modeling papers (e.g., Kirov and Cotterell 2018) discard CELEX lemma frequency and the SIGMORPHON shared tasks except for 2017 and 2022 sampled data uniformly rather than weighted by frequency. We expect that weighted sampling should have a greater effect in cases where irregular items are

<sup>3</sup>Kirov and Cotterell (2018) report *u*-shaped learning for English, however on close inspection, it is not the phenomenon described in the developmental literature. They clarify (§2.1, 5.1) that is is “*micro u*-shaped learning,” which is really small oscillations in the performance curve over training epochs with a fixed vocabulary.

<sup>4</sup>Estimating learner vocabulary knowledge is notoriously challenging. One major issue is the distinction between expressive or production vocabulary on one hand and receptive or comprehension vocabulary on the other. Another issue is whether the unit being counted is lemmas (or “base words”) or inflected words, which naturally provide more divergent numbers for languages with larger morphological paradigms. A three-year-old’s entire production vocabulary in base words is measured in the hundreds (Fenson et al., 1994; Szagun et al., 2006; Hart and Risley, 2003) with substantial individual variation. We are unaware of raw number estimates for a 3-year-old’s receptive vocabulary, though (Fenson et al., 1994) discusses a mean of 150-200 words for 16-month-olds.

Our data only consists of one part of speech per language. Naturally this means only a fraction of the vocabulary is directly relevant for our experiments. Keeping this in mind, along with the large variance introduced by individual variation, “hundreds” is the appropriate order of magnitude for the lexicon size of our model learners.

skewed towards the high end of the frequency distribution. For smaller samples, these irregular types will dominate the training data when weighting sampling is employed but not when uniform sampling is.

We do not evaluate a single train-test split per language, as is typically done in this area. We make and compare five splits with unique random seeds. As well as a generally reasonable approach in NLP model evaluation (Gorman and Bedrick, 2019), this allows us to investigate the stability of performance on different data sets. Random sampling has been shown to significantly affect performance in a variety of morphology learning tasks (Liu and Prud’hommeaux, 2022; Kodner et al., 2023b), so it is likely to have an effect here too. Developmentally, this question corresponds to the well-known observation in child language acquisition that individual learning trajectories are relatively homogeneous despite the fact that each child receives their own unique input (i.e., their own random sample; Labov 1972). Thus we expect a cognitively plausible model to be relatively resilient to which items happen to occur or not occur in its training data.

## 3 Experiments

### 3.1 Data sources and preparation

We investigated three morphological patterns, all of which have been previously studied from developmental and computational perspectives: English past tense inflection, German noun pluralization, and Arabic noun pluralization. These phenomena are ideal for this task because ANN learning trajectories and error patterns can be compared against existing developmental literature (e.g., Ravid and Farah, 1999; Clahsen and Rothweiler, 1993; Marcus et al., 1992, 1995)

Unsplit data was taken from the 2022 SIGMORPHON developmental sub-task and is all orthographic (Arabic is diacritized).<sup>5</sup> English and German lemmas were extracted from CHILDES (MacWhinney, 2000) child-directed speech (CDS) and intersected with UniMorph (Batsuren et al., 2022) to remove errorful annotations; we then converted them to a standard format. Extracting forms from CDS provided frequency estimates for typical morphological input during acquisition and removed rare and unusual items from UniMorph. Frequencies for Arabic were extracted from the Penn Arabic Treebank (Maamouri et al., 2004) because the Arabic corpora in CHILDES lack the necessary morphological annotations for this task.

Following the format widely adopted for morphological inflection tasks in recent years, training data was presented as a list of (**lemma**, **inflected form**, **morpho-syntactic feature set**) triple types, and test types were presented as (**lemma**, **feature set**) pairs, where the learner is asked to provide the appropriate inflected form. This is effectively the computational adaptation of the classic Wug test paradigm following Berko (1958). (1)-(2) provide example English and German training and test items.

| English Training Items |              |                               | English Test Items |                           |  |
|------------------------|--------------|-------------------------------|--------------------|---------------------------|--|
| (1)                    | <b>run</b>   | <b>ran</b> <b>V;PST</b>       | <b>see</b>         | <b>?</b> <b>V;PST</b>     |  |
|                        | <b>walk</b>  | <b>walked</b> <b>V;PST</b>    | <b>look</b>        | <b>?</b> <b>V;PST</b>     |  |
| German Training Items  |              |                               | German Test Items  |                           |  |
| (2)                    | <b>Lampe</b> | <b>Lampen</b> <b>N;FEM;PL</b> | <b>Paar</b>        | <b>?</b> <b>N;NEUT;PL</b> |  |
|                        | <b>Tanz</b>  | <b>Tänze</b> <b>N;MASC;PL</b> | <b>Amse1</b>       | <b>?</b> <b>N;FEM;PL</b>  |  |

### 3.2 Data splits

We split the original data to generate training, fine-tuning, development, and test splits. This was repeated five times with unique random seeds to produce distinct data sets. The training+fine-tuning sets were sub-sampled into nested sets of size 100, 200, and so on in order to approximate a learning curve as vocabulary size increases. These sets were split uniformly at random into 80% training and 20% fine-tuning. The maximum English and Arabic sets contained 1000 items, while the maximum German sets were restricted to 600 due to limitations of the original data set. All development sets contained 500 items. Remaining

<sup>5</sup>The input to children is of course phonological and not orthographic. While prior work on this particular inflection task has shown little effect of presentation style (orthography vs. phonological transcriptions; Kodner et al. 2023a), style has been shown to have an effect in other tasks (Gale et al., 2023).

items were assigned to test (Arabic: 496, English: 554, German: 600). Dev and test sets were kept consistent regardless of training size. No lemmas overlapped between the sets. Two sampling strategies were adopted:

**UNIFORM:** As in most prior work, data was partitioned uniformly at random without replacement. This ignored any information about frequency of the triples or lemmas. An advantage of this approach is that it can be performed on data sets with no frequency information, but it does not reflect the situation in language acquisition, where frequent forms are more likely to appear early in a learner’s input (e.g., Goodman et al., 2008).

**WEIGHTED:** Sub-sampling was weighted by frequency without replacement. The first 100 training+fine-tuning set was sampled from the entire data set, followed by 100 more items to create the 200 training+fine-tuning set, and so on. After training was completely sampled, dev+test were sampled together from the remainder and then split uniformly at random. Thus, the smallest training sets are skewed towards the highest-frequency items and dev+test are skewed towards lower frequency items. We take this to reflect the situation during language acquisition: high-frequency items are more likely to be present in a typical child’s input and thus can be memorized, while low-frequency items are less likely to be presented to any given child early on and will often need to be inferred from morphological knowledge constructed over mostly higher-frequency forms.

We expected the choice of sampling strategy to be impactful for English verbs in particular, since it is clear in the data that the irregular verbs are heavily skewed towards the high end of the frequency range. For lower training sizes, irregulars should make up an especially large proportion of training items, which should manifest in lower test accuracy compared to UNIFORM.

### 3.3 Evaluated systems

We evaluated three artificial neural network models (ANNs). Two, CHR-TRM and CLUZH, were chosen for their availability and performance on recent SIGMORPHON shared tasks. E-D was chosen because it positioned itself specifically as a cognitive model of morphology acquisition.

CHR-TRM (Wu et al., 2019) is a character-level transformer that enforces strict monotonicity between the input and the output strings; the model jointly learns the transduction and the monotonic alignment process (Wu and Cotterell, 2019), previously holding the state-of-the-art in SIGMORPHON task and serving as a baseline in 2021 and 2022. We used the provided default hyper-parameters for small training conditions.

CLUZH (Wehrli et al., 2022) is a character-level neural transducer that employs a bidirectional long-short term memory (LSTM) encoder-decoder architecture; for each input string, this model converts it to the output string via performing character edits including deletion or insertion. This model also has a copying mechanism, directly copying characters from input to output (Makarov and Clematide, 2018); it implements teacher forcing so that the model does not have access to its own predictions during training, which is presumed to help with the generalizability of the model; lastly, this model employs mini-batching for training, resulting in more efficient computation; all these aforementioned traits set CLUZH apart from CHR-TRM, possibly making the former more adapted to data limitation scenarios, which is the case with our study. For the 2022 SIGMORPHON share tasks, CLUZH outperformed CHR-TRM. While the SIGMORPHON submission was optimized for each language individually, we used consistent hyper-parameters to facilitate comparison. We analyzed two different decoding methods, **cluzh-b4** with beam decoding, size=4, and **cluzh-gr** with greedy decoding.

E-D (Kirov and Cotterell, 2018) is an LSTM-based sequence-to-sequence model with the encoder and the decoder each having 300 hidden units. This ANN was originally framed as support for the connectionist side in a renewed Past Tense Debate, and is thus taking an explicit stance on its cognitive plausibility (cf. Rumelhart and McClelland, 1986; Pinker and Prince, 1988).

## 4 Quantitative Analysis

In this section, we present quantitative analyses of model performance. Throughout this section, we report results on the test set; we carried out the same series of examinations for the dev set and there was no observable qualitative difference in the findings. All reported accuracies measure the rate of exact matches between predicted and target forms.

## 4.1 Effect of training size

We analyzed the effect of different training sizes on the overall accuracy, summarized in Tables 1 and 2. After calculating performance across data splits under the two sampling strategies (UNIFORM and WEIGHTED) for all languages, we employed a mixed effect model which predicts the overall accuracy score (between 0 and 1) on the test data as a function of the training size, controlling for the language, the model type, and the sampling strategy, along with interactions between each of the aforementioned four fixed effects, expressed in R syntax in (3). Statistics were performed with R; full descriptions of our models and their outputs are provided in Appendix A.

$$(3) \quad \text{ACCURACY} \sim \text{LANG} * \text{MODEL} * \text{TRAIN SIZE} * \text{SAMPLING STRATEGY}$$

Summarizing the main findings of the regression model, there is a significant effect of training size overall ( $\beta = 9.702\text{e-}04$ ,  $p = 2.01\text{e-}09$ ), which may be interpreted as predicting a roughly 49-point performance gain as training size increases by 500 and an 87-point gain as training size increases by 900. This is consistent with visual inspection of Figures 1 and 2: Performance improves across all models, languages, and seeds as training size increases, though this improvement is steeper for some models than others. CHR-TRM ( $\beta = 1.167\text{e-}03$ ,  $p = 1.67\text{e-}06$ ) and E-D ( $\beta = 1.224\text{e-}03$ ,  $p = 8.16\text{e-}07$ ) are particularly sensitive to training size, consistently far under-performing the CLUZH models at small training sizes. CLUZH’s architecture was optimized for exactly this low-training scenario.

| <b>Arabic</b>  | CHR-TRM       | CLUZH-b4      | CLUZH-gr      | E-D           |
|----------------|---------------|---------------|---------------|---------------|
| 100            | 6.61 (3.02)   | 42.14 (11.90) | 41.01 (13.91) | 1.41 (3.23)   |
| 200            | 29.76 (7.46)  | 50.36 (12.70) | 49.19 (12.70) | 18.79 (3.83)  |
| 300            | 45.12 (8.06)  | 55.65 (7.46)  | 54.76 (7.46)  | 30.97 (6.45)  |
| 400            | 50.77 (10.28) | 58.79 (3.63)  | 57.66 (5.65)  | 39.31 (7.86)  |
| 500            | 53.83 (3.02)  | 58.99 (8.47)  | 57.66 (8.47)  | 42.42 (6.45)  |
| 600            | 57.62 (3.43)  | 63.47 (7.06)  | 62.46 (7.26)  | 46.85 (6.85)  |
| 700            | 58.83 (4.44)  | 62.22 (5.04)  | 60.48 (6.45)  | 48.63 (7.66)  |
| 800            | 60.20 (5.24)  | 64.68 (3.43)  | 64.11 (3.43)  | 50.56 (3.43)  |
| 900            | 63.39 (4.64)  | 65.56 (4.03)  | 64.23 (2.22)  | 51.13 (5.65)  |
| 1000           | 61.81 (8.47)  | 66.25 (8.47)  | 65.65 (7.26)  | 51.45 (7.46)  |
| <b>German</b>  | CHR-TRM       | CLUZH-b4      | CLUZH-gr      | E-D           |
| 100            | 12.87 (7.50)  | 69.60 (8.17)  | 70.17 (5.00)  | 3.67 (15.83)  |
| 200            | 42.87 (7.33)  | 73.80 (2.50)  | 73.80 (2.50)  | 56.47 (4.17)  |
| 300            | 61.57 (11.00) | 75.30 (4.67)  | 75.13 (4.83)  | 60.77 (6.00)  |
| 400            | 67.90 (4.00)  | 76.73 (1.83)  | 76.83 (1.83)  | 66.30 (3.67)  |
| 500            | 71.37 (3.33)  | 78.17 (4.50)  | 77.93 (4.50)  | 67.67 (6.33)  |
| 600            | 73.93 (3.00)  | 78.13 (4.33)  | 78.23 (4.50)  | 70.03 (4.83)  |
| <b>English</b> | CHR-TRM       | CLUZH-b4      | CLUZH-gr      | E-D           |
| 100            | 25.02 (9.39)  | 81.08 (3.97)  | 81.12 (3.97)  | 32.31 (16.61) |
| 200            | 55.45 (9.03)  | 82.56 (6.14)  | 82.67 (6.32)  | 68.70 (2.71)  |
| 300            | 71.88 (2.35)  | 83.21 (3.79)  | 83.25 (4.15)  | 78.92 (7.40)  |
| 400            | 77.29 (7.58)  | 84.80 (8.12)  | 84.66 (8.12)  | 81.84 (5.78)  |
| 500            | 81.48 (5.05)  | 85.09 (5.42)  | 85.09 (5.60)  | 84.15 (4.69)  |
| 600            | 83.94 (7.40)  | 86.32 (5.05)  | 86.14 (5.42)  | 84.87 (6.50)  |
| 700            | 85.05 (5.96)  | 87.94 (2.71)  | 87.87 (3.07)  | 86.43 (3.97)  |
| 800            | 85.78 (5.60)  | 87.22 (6.32)  | 87.26 (6.50)  | 86.61 (3.43)  |
| 900            | 85.96 (5.23)  | 88.38 (6.50)  | 88.23 (6.14)  | 87.73 (6.68)  |
| 1000           | 86.71 (5.60)  | 88.77 (4.69)  | 88.52 (4.87)  | 87.33 (7.40)  |

Table 1: UNIFORM: Mean percent accuracy across seeds and score range (*italicized*) across seeds by language and ANN.

## 4.2 Effect of sampling strategy

The effect of sampling strategy (UNIFORM vs WEIGHTED) on performance is summarized in Tables 1 and 2 and in Figures 1 and 2. The performance curves are visually similar across sampling strategy, and indeed, strategy came out as insignificant in our model. However, across all languages, ANNs, training sizes, and seeds, average performance is slightly higher for UNIFORM (65.59%) than for WEIGHTED (62.79%).

| <b>Arabic</b>  | CHR-TRM                | CLUZH-b4               | CLUZH-gr               | E-D                    |
|----------------|------------------------|------------------------|------------------------|------------------------|
| 100            | 4.72 ( <i>3.83</i> )   | 41.77 ( <i>12.10</i> ) | 38.55 ( <i>15.73</i> ) | 0.65 ( <i>0.60</i> )   |
| 200            | 25.24 ( <i>9.88</i> )  | 51.21 ( <i>4.84</i> )  | 49.92 ( <i>4.84</i> )  | 12.50 ( <i>10.28</i> ) |
| 300            | 37.86 ( <i>12.90</i> ) | 57.38 ( <i>5.44</i> )  | 56.25 ( <i>5.65</i> )  | 25.20 ( <i>12.70</i> ) |
| 400            | 47.98 ( <i>10.69</i> ) | 58.51 ( <i>4.44</i> )  | 57.42 ( <i>5.24</i> )  | 35.73 ( <i>2.42</i> )  |
| 500            | 54.31 ( <i>5.65</i> )  | 61.41 ( <i>3.02</i> )  | 60.93 ( <i>5.04</i> )  | 42.90 ( <i>7.66</i> )  |
| 600            | 56.41 ( <i>7.86</i> )  | 61.37 ( <i>8.67</i> )  | 60.08 ( <i>7.86</i> )  | 45.48 ( <i>7.86</i> )  |
| 700            | 57.82 ( <i>9.07</i> )  | 63.19 ( <i>5.44</i> )  | 62.02 ( <i>7.06</i> )  | 47.30 ( <i>6.65</i> )  |
| 800            | 59.60 ( <i>3.83</i> )  | 65.24 ( <i>5.44</i> )  | 64.35 ( <i>4.03</i> )  | 50.93 ( <i>4.64</i> )  |
| 900            | 61.57 ( <i>4.44</i> )  | 64.48 ( <i>4.84</i> )  | 63.79 ( <i>4.44</i> )  | 52.14 ( <i>10.48</i> ) |
| 1000           | 64.48 ( <i>6.25</i> )  | 66.09 ( <i>5.24</i> )  | 65.28 ( <i>4.23</i> )  | 53.79 ( <i>6.45</i> )  |
| <b>German</b>  | CHR-TRM                | CLUZH-b4               | CLUZH-gr               | E-D                    |
| 100            | 8.77 ( <i>7.33</i> )   | 61.57 ( <i>25.17</i> ) | 61.87 ( <i>22.67</i> ) | 7.63 ( <i>20.33</i> )  |
| 200            | 35.27 ( <i>10.83</i> ) | 73.27 ( <i>7.67</i> )  | 73.23 ( <i>8.17</i> )  | 45.30 ( <i>24.83</i> ) |
| 300            | 56.03 ( <i>6.50</i> )  | 75.47 ( <i>3.50</i> )  | 75.07 ( <i>3.50</i> )  | 57.87 ( <i>3.67</i> )  |
| 400            | 62.23 ( <i>3.83</i> )  | 76.07 ( <i>3.83</i> )  | 75.87 ( <i>4.67</i> )  | 60.37 ( <i>6.83</i> )  |
| 500            | 69.00 ( <i>5.17</i> )  | 78.03 ( <i>2.67</i> )  | 77.87 ( <i>3.50</i> )  | 64.73 ( <i>2.83</i> )  |
| 600            | 70.60 ( <i>6.00</i> )  | 78.63 ( <i>5.17</i> )  | 78.67 ( <i>4.83</i> )  | 66.30 ( <i>3.67</i> )  |
| <b>English</b> | CHR-TRM                | CLUZH-b4               | CLUZH-gr               | E-D                    |
| 100            | 8.77 ( <i>4.51</i> )   | 78.27 ( <i>8.30</i> )  | 78.59 ( <i>8.30</i> )  | 8.23 ( <i>7.22</i> )   |
| 200            | 30.32 ( <i>6.32</i> )  | 83.57 ( <i>4.87</i> )  | 83.86 ( <i>4.15</i> )  | 36.97 ( <i>10.29</i> ) |
| 300            | 45.67 ( <i>5.05</i> )  | 82.42 ( <i>5.05</i> )  | 82.06 ( <i>3.79</i> )  | 59.35 ( <i>14.26</i> ) |
| 400            | 61.19 ( <i>9.75</i> )  | 84.44 ( <i>3.61</i> )  | 84.26 ( <i>4.15</i> )  | 69.75 ( <i>5.23</i> )  |
| 500            | 74.66 ( <i>3.61</i> )  | 85.60 ( <i>4.87</i> )  | 85.31 ( <i>4.69</i> )  | 78.99 ( <i>5.78</i> )  |
| 600            | 78.63 ( <i>4.87</i> )  | 87.08 ( <i>3.25</i> )  | 86.82 ( <i>3.25</i> )  | 81.77 ( <i>3.97</i> )  |
| 700            | 82.56 ( <i>7.94</i> )  | 87.80 ( <i>3.61</i> )  | 87.69 ( <i>3.79</i> )  | 85.02 ( <i>3.07</i> )  |
| 800            | 84.08 ( <i>1.99</i> )  | 88.34 ( <i>2.53</i> )  | 88.34 ( <i>2.71</i> )  | 85.16 ( <i>5.23</i> )  |
| 900            | 86.93 ( <i>3.97</i> )  | 89.42 ( <i>5.05</i> )  | 89.24 ( <i>4.87</i> )  | 87.18 ( <i>2.35</i> )  |
| 1000           | 87.33 ( <i>5.23</i> )  | 90.32 ( <i>3.43</i> )  | 90.11 ( <i>2.71</i> )  | 88.81 ( <i>2.17</i> )  |

Table 2: WEIGHTED: Mean percent accuracy across seeds and score range (*italicized*) across seeds by language and ANN.

Sampling strategy only had a significant effect in one case: the performance of CHR-TRM ( $\beta = -7.806\text{e-}01$ ,  $p = 0.018$ ) and E-D ( $\beta = -8.310\text{e-}01$ ,  $p = 0.014$ ) on English. The learning curves of both architectures on English improved much more steeply at smaller training sizes when UNIFORM sampling is applied than when WEIGHTED is applied. This is expected given the observation, discussed further in Section 5.3, that irregular verbs in English are heavily skewed towards the higher frequency range. WEIGHTED sampling creates unfavorable early training samples regardless of seed choice and thus a more gradual learning curve. The important conclusion from this analysis is that, while sampling strategy makes little difference in performance at larger training sizes, it does make a difference at smaller developmentally-plausible training sizes for some morphological phenomena. Full descriptions and outputs for our regressions can be found in Appendix A.

### 4.3 Variation across random seeds

We concluded our quantitative analysis with a look into the variability introduced by random sampling. We calculated two metrics, which we refer to as *score range*, summarized in Tables 1 and 2, and *random seed variability*. The former measures the difference between the lowest and the highest accuracy across seeds, while the latter computes the standard deviation of the accuracy scores across the seeds.

We first analyzed the score variation across random seeds for each language. Across sampling strategy, training sizes and ANNs, we found that both average score ranges (Arabic: 6.65%, German: 6.74%, English: 5.44%) and average random seed variability (Arabic: 2.66%, German: 2.73%, English: 2.17%) are fairly consistent across languages. There is similarly little difference between sampling strategies for average score ranges (UNIFORM: 6.01%, WEIGHTED: 6.39%) or random seed variability (UNIFORM: 2.39%, WEIGHTED: 2.58%). Altogether, these results are reassuring in that results obtained from one train-test sample seem to be representative of typical outcomes.

Finally, we performed two regressions for score range and random seed variability respectively to study the relationship between training size and score variation. For both models, we included the training size as a fixed effect, controlling for the effects of language, model types, and sampling strategies, with interaction terms between all aforementioned fixed effects, as in (4).

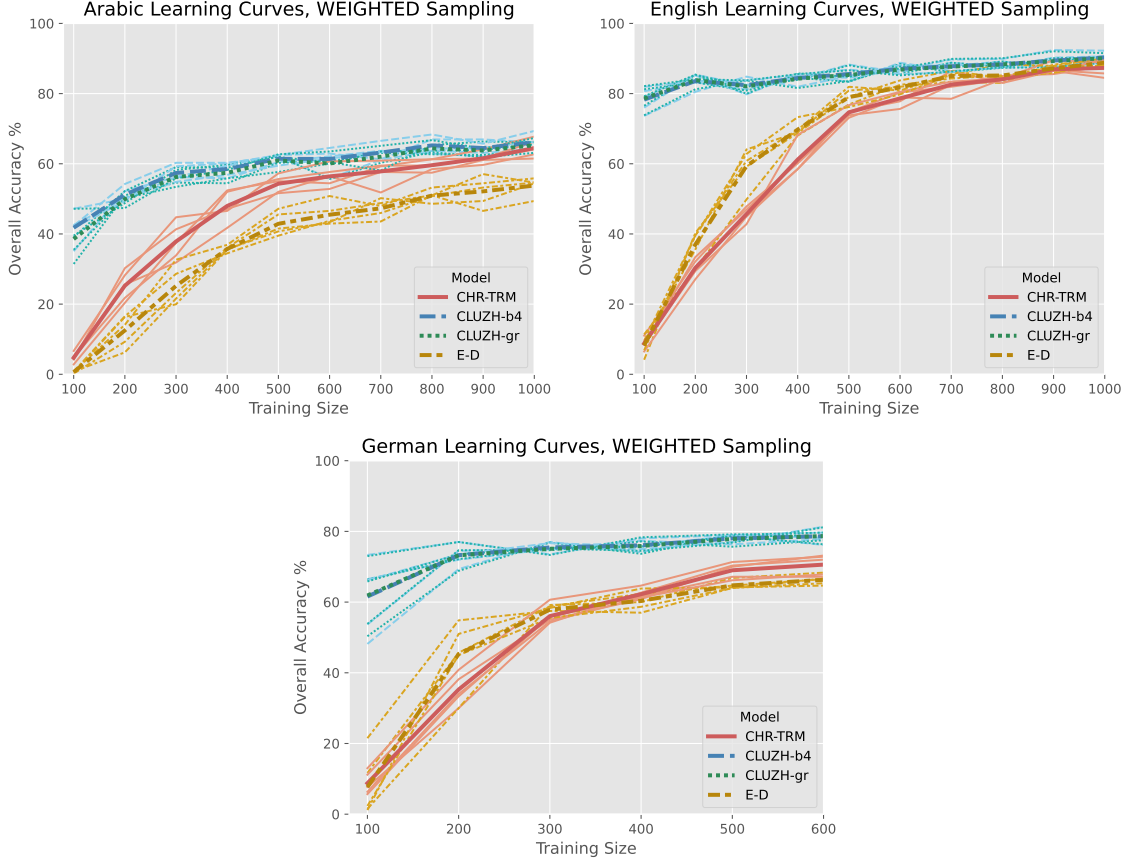


Figure 1: Learning curves on WEIGHTED data splits for Arabic nouns, German nouns, and English verbs. Thin/light lines = individual seeds, and thick/dark lines = averages across seeds.

$$(4) \text{ SCORE RANGE/SEED VARIABILITY} \sim \text{LANG} * \text{MODEL} * \text{TRAIN SIZE} * \text{SAMPLING STRAT}$$

Training size had a small but significant negative effect on both score range ( $\beta = -2.019\text{e-}04$ ,  $p = 0.003$ ) and random seed variability ( $\beta = -9.434\text{e-}05$ ,  $p = 0.007$ ), indicating that variability decreased as training size increased. This follows from the intuition that the presence of any given item in training or test has a lesser impact on a model as the number of training items increases. The interaction between training size and sampling strategy is not significant in either case. Full outputs of our regression model can be found in Appendix B.

## 5 Linguistic Analysis

For a model to be cognitively plausible, it should not only achieve high performance but also learn in a similar way to children. This section evaluates the ANNs’ outputs in terms of how well they reproduce observed patterns in Arabic, German, and English child language acquisition. For consistency, all annotations are performed on predictions from WEIGHTED seed 0.

### 5.1 Arabic Noun Pluralization

Arabic forms noun plurals in two ways: by suffixation (*sound plurals*) or by stem mutation (*broken plurals*). There are two sound plural suffixes (ignoring case and state marking), feminine (FEM) *-āt*, and masculine (MASC) *-ūn*. The relationship between gender and sound plural ending is generally reliable, but some MASC nouns, generally non-human, take the FEM sound plural. Broken plurals can be divided into many subclasses

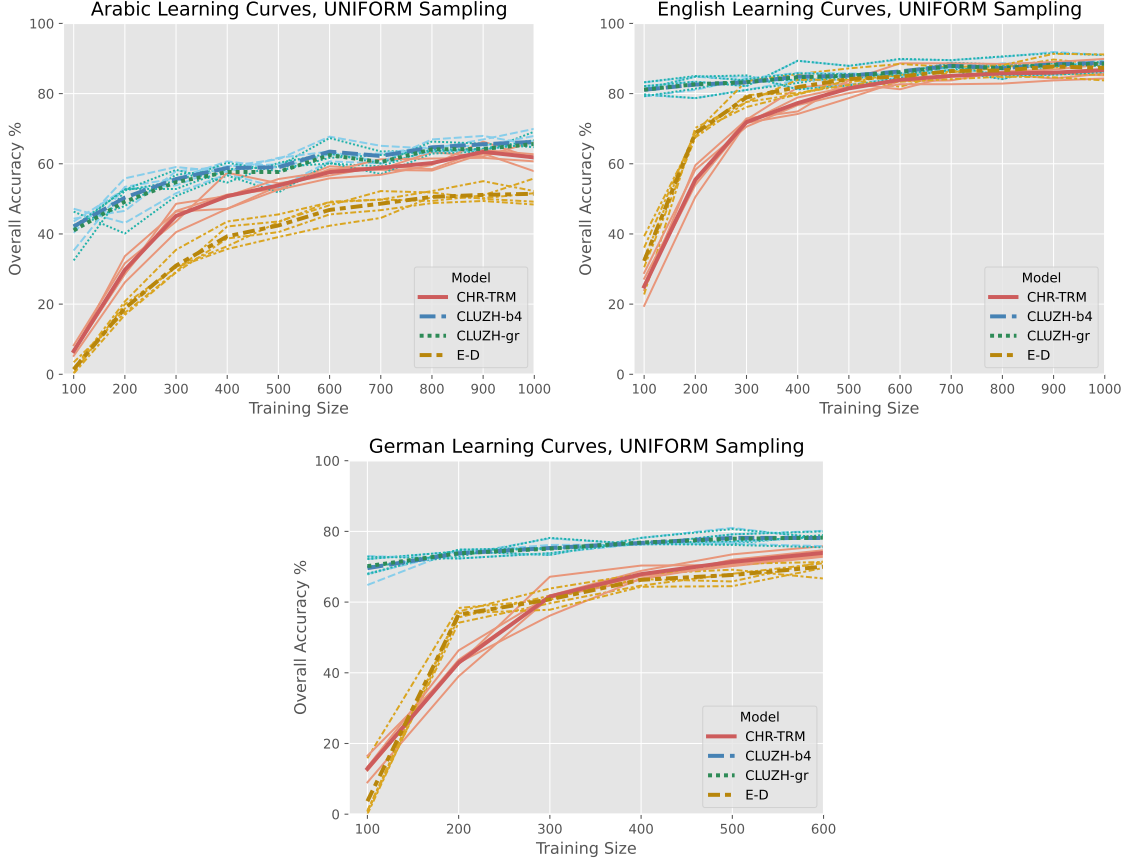


Figure 2: Learning curves on UNIFORM data splits for Arabic nouns, German nouns, and English verbs. Thin/light lines = individual seeds, and thick/dark lines = averages across seeds.

according to which templatic pattern defines the output of their stem mutations. In Modern Standard Arabic (MSA), there are approximately 30 broken plural patterns (McCarthy and Prince, 1990), though the exact count depends on the level of abstraction assumed for the templatic pattern.

Ravid and Farah (1999) identified two kinds of *u*-shaped learning in Palestinian Arabic-learning children: First, they began by accurately distinguishing MASC and FEM sound plurals followed by over-application of the FEM to MASC forms before returning to high accuracy. Second, they also went through a period of over-applying FEM sound plurals to what should be broken plurals. On the other hand, they found very few instances of FEM-to-MASC sound and broken-to-broken errors for most age groups.

We compared E-D and CHR-TRM to Ravid and Farah (1999) directly by evaluating their ability to inflect known words from training. Only E-D produced more than one error across all seeds and training sizes. We annotated the predictions of E-D on known words for training sizes 200, 400, 600, 800, and 1,000 (seed 0), in order to identify whether it reproduced attested *u*-shaped learning. The annotations make a distinction between the two sound plurals (FEM and MASC) to detect sound-to-sound errors (**Snd**→**Snd**), while broken plurals are annotated as **Br** regardless of the plural pattern. This level of granularity is adopted from Dawdy-Hesterberg and Pierrehumbert (2014), which tested several non-neural computational models for Arabic morphology acquisition.

The system was highly accurate, making only 54 errors across these training sizes, many of which were repeated. There were no nonsense errors, and the large majority were sound-to-sound errors with swapped gender suffixes on human nouns like ‘pilot’ and ‘Palestinian,’ in cases where both forms with a gender feature tag were present in the data. It appears as though E-D did not learn to distinguish the inflection for masculine/feminine pairs when both were provided. On a given training size, it would prefer either masculine or feminine inflection.





Figure 3: Breakdown of E-D’s Arabic errors on known words across training sizes. ? indicates nonsense outputs, which are absent.

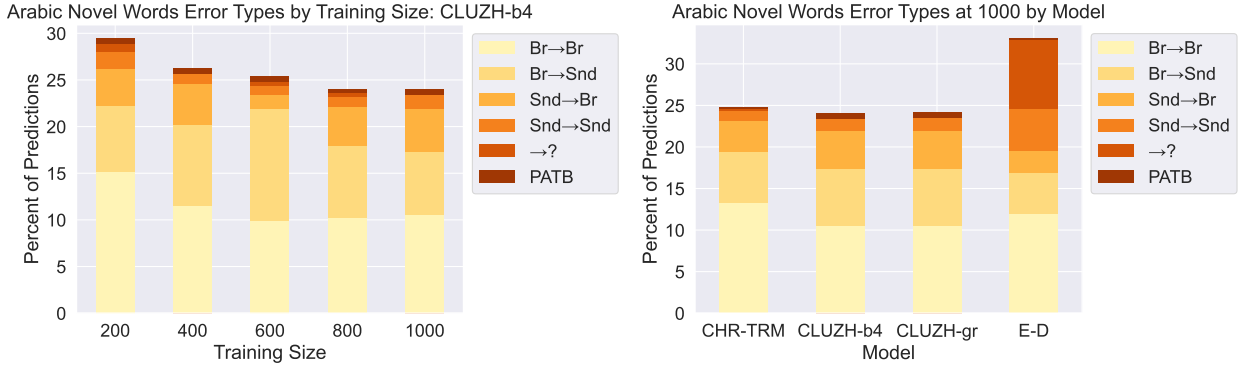


Figure 4: Top: Breakdown of CLUZH-B4’s Arabic errors on novel words across training sizes. Bottom: Breakdown of each neural model’s Arabic errors at maximum training. PATB indicates errors attributable to annotation mistakes in the original PATB data. ? indicates nonsense outputs.

Excluding these gender-swap paired case left 22 errors (0.73% of all predictions on the training data), which are visualized in Figure 3. While the number of  $\text{Snd} \rightarrow \text{Snd}$  and  $\text{Br} \rightarrow \text{Snd}$  errors increased as training size increased as in Ravid and Farah (1999), they occur on training items that were absent from smaller training sizes. In other words, the model does not begin to over-apply these patterns on words that it had predicted correctly on smaller training sizes. Of the ten total  $\text{Snd} \rightarrow \text{Snd}$  errors, seven are MASC to FEM, which does reproduce the developmentally expected asymmetry.

Next, we applied the same annotation scheme to the full test set. We evaluate each ANN once trained on the full 1,000 training items as well as CLUZH-B4, the best performing model, on a range of training sizes to evaluate its learning trajectory. The results are provided in Figure 4, in the style of Dawdy-Hesterberg and Pierrehumbert (2014). We made some key observations. First, learning was monotonic. Neither type of  $u$ -shaped learning was observed. Second,  $\text{Br} \rightarrow \text{Snd}$  errors were indeed relatively common, as they are in children. Third, however, most errors were  $\text{Br} \rightarrow \text{Br}$  (e.g., *nabiyy-<sup>\*</sup>nab* instead of *nabiyy-ʔanbiyāʔ*) or  $\text{Snd} \rightarrow \text{Br}$  (e.g., *maḡhūd-<sup>\*</sup>maḡāhīd* instead of *maḡhūd-maḡhūdāt*). These are less common developmentally. Across all annotated predictions, 32  $\text{Snd} \rightarrow \text{Snd}$  were MASC  $\rightarrow$  FEM, while 20 were the reverse. However, FEM  $\rightarrow$  MASC errors were proportionately far more common than what is attested developmentally.

An error analysis of CLUZH-GR, CHR-TRM and E-D uncovers qualitatively similar patterns. Overall, the ANNs were highly accurate on known words. Their errors on novel words did not apparently correspond to children’s errors.

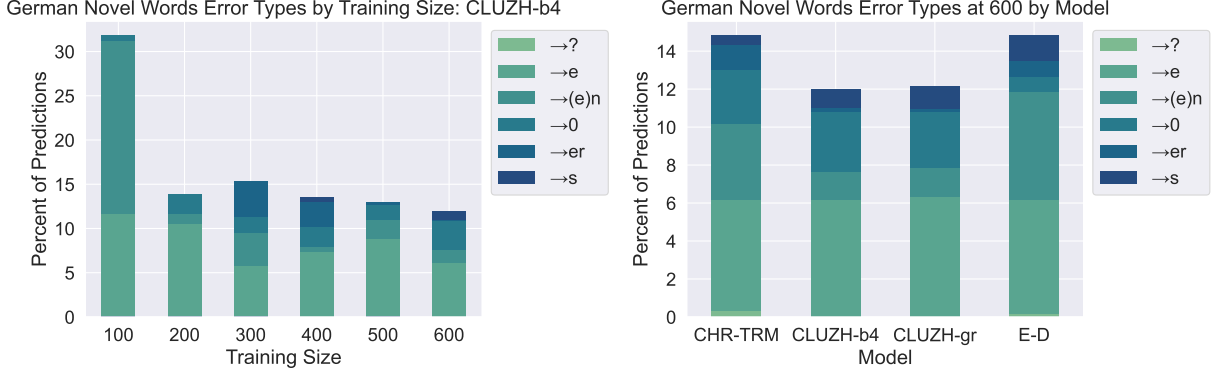


Figure 5: Top: Breakdown of CLUZH-B4’s German errors on novel words across training sizes. Bottom: Breakdown of each neural model’s German errors at maximum training. ? indicates nonsense outputs.

## 5.2 German Noun Pluralization

German nouns are inflected with one of five (nominative and accusative) plural patterns:  $-(e)n$ , the most frequent, especially among FEM nouns,  $-e$ , the second most frequent,  $-\emptyset$ ,  $-er$ , and  $-s$ , the least frequent. This is interesting because  $-s$  appears to be the default form of last resort despite its low frequency (Marcus et al., 1995). Thus, German noun plurals provide a way to distinguish productivity from frequency.

E-D and CHR-TRM were highly accurate when evaluated on known words from their training data, with 100% accuracy achieved on nearly all training sizes and seeds. All errors that did emerge were nonsense errors, for example, E-D produced *\*Spritzische* for *Spritzfische* on 400 train seed 1. Of these, only one error involved a suffix, though it could be analyzed as part of a nonsense error: E-D produced *\*Einwupnere* for *Einwohner* on 500 train seed 2. There was no evidence of  $-s$  generalization. Furthermore, this represents a super-human level of performance. We do not expect learners to master inflection, even on known words, when their vocabularies only contain 100 verbs.

Test plurals were annotated to indicate which of the five pluralization types was applied:  $-e$ ,  $-s$ ,  $-er$ ,  $-(e)n$ , or  $-\emptyset$ . Since the goal was not to perfectly capture German orthography, failure to capture consonant doubling (*-inen* instead of expected *-innen*) was not penalized. Model predictions that did not fit into any of these categories (usually due to unnatural stem-internal changes) were labeled ?.

Figure 5 breaks down error types by training size on test words for CLUZH-B4, the overall best performing ANN, as well as the final predictions for each ANN. At 200 and above, the over-application of  $-e$  was consistently the most common error type. Early over-application of  $-e$  is consistent with child development, where  $-e$  and  $-\emptyset$  are acquired very early (Gawlitze-Maiwald, 1994).

$-(e)n$  is the default ending for FEM nouns. Indeed CLUZH-B4 applied it to FEM near-categorically and also over-extended it to MASC and neuter NEUT nouns. It over-extended the  $-s$  plural for a time, particularly around 300-400 items in training, which is consistent with developmental expectations (Elsen, 2002). Figure 5 shows each neural model’s error types at maximum training on seed 0. For each of the models, over-application of  $-e$  was dominant followed by  $-(e)n$  and  $-\emptyset$ . The error rate was unacceptably high, but learning trajectories on unseen data were roughly what would be expected from a cognitively-plausible learning model.

## 5.3 English Past Tense

For English, we began with an investigation of the rate of irregulars in the training data derived from UNIFORM vs. WEIGHTED sampling, to confirm the cause of the slower WEIGHTED performance curves uncovered in Section 4.1. Figure 6 plots the proportion of irregulars in each training set, which we define as those verbs whose past tense forms involve any change other than suffixing  $-(e)d$  with doubling of the final orthographic consonant as required. Two patterns stand out. First, WEIGHTED sampling yielded training sets with higher rates of irregulars, consistent with the fact that irregulars are skewed towards the high end of the frequency range. Second, while the rate of irregulars was flat for UNIFORM, it was highest for

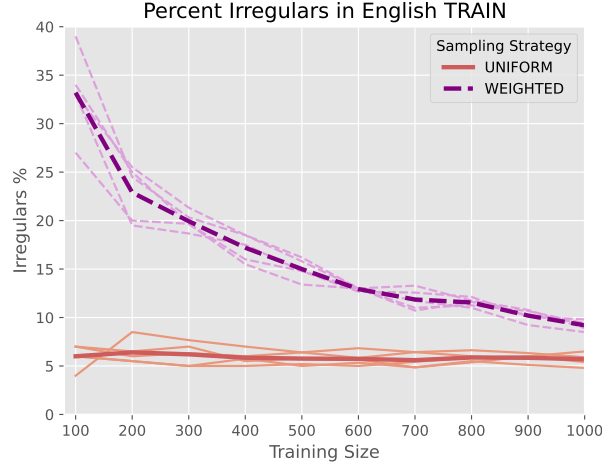


Figure 6: Rate of irregular verbs in English WEIGHTED and UNIFORM training sets. Thin/light lines = individual seeds, and thick/dark lines = averages across seeds.

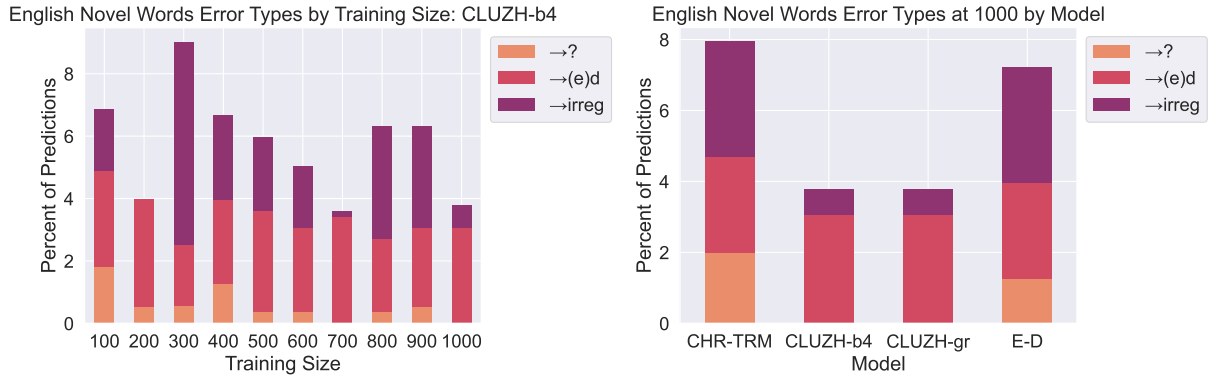


Figure 7: Top: Breakdown of CLUZH-B4’s English errors on novel words across training sizes. Bottom: Breakdown of each neural model’s English errors at maximum training. ? indicates nonsense outputs.

WEIGHTED when training sizes are smaller. In both cases, WEIGHTED was more consistent with the input during language acquisition. This further explains why overall performance on WEIGHTED was slightly lower for CHR-TRM and E-D. The higher proportion of irregulars makes for a more challenging learning problem, especially at smaller training sizes.

We found that accuracy for known words on the training data was perfect or near perfect for both CHR-TRM and E-D on all seeds and training sizes. The sole interesting error cases came from CHR-TRM, which achieved only 10% accuracy at 100 training (all nonsense errors) as well as one error (null marking) at 600 training on seed 0. It also made three errors at 700 training on seed 2. These last three errors could be classified: there was a clear instance of over-regularization (*\*falled* for *fell*), one instance of over-regularization plus unusual character doubling (*\*goed* for *went*), and one instance of over-irregularization, specifically a stem vowel change with no suffix (*\*trot* for *treated*).

Test accuracy was quite a bit lower, so we followed an equivalent procedure for English as for German and Arabic, this time annotating three inflection types: **-(e)d** (regular, e.g., *smear-smear*ed), **irreg** (irregular or analogized to irregular, e.g., *fly-flew*, *tweet-tweet*ed), and **?** (unnatural predictions, e.g., *correspond-correspond*). Because the goal of this evaluation was not to capture the idiosyncrasies of English orthography, near-correct regular predictions such as *obey-obeid* and *trim-trimed* were annotated as **-(e)d**.

The results are displayed in Figure 7. On the full training set, both CLUZH-GR and CLUZH-B4 made only over-regularization and over-irregularization errors, with over-regularizations dominating. However, the CLUZH systems showed much less of an asymmetry between over-regularization and over-irregularizations than English learning children, for whom over-regularizations constitute 97.8% of all production errors (Marcus et al., 1992; Maslen et al., 2004; Xu and Pinker, 1995) and who strongly favor the application of regular *-(e)d* over stem changes to novel words (cf. Berko, 1958). By contrast, over-regularizations were in the minority for CHR-TRM and E-D, which produced a greater proportion of unnatural errors and over-irregularizations.

No *u*-shaped learning pattern of over-regularization emerged from any of the models on known words, even though this is expected developmentally for English learners (Marcus et al., 1992; Prasada and Pinker, 1993). In fact, all models achieved perfect performance across nearly all seeds even on only 100 training items. This constitutes super-human performance, as for German, since a child who knows 100 verbs is not expected to even consistently apply *-(e)d* (Fenson et al., 1994; Marcus et al., 1992). A further analysis of CLUZH-B4 on test words (Figure 7) finds no *u*-shaped pattern of over-regularization on novel words either, though it is not clear if one should be expected. Overall, the systems are not particularly human-like, even though CLUZH at first appears accurate on some training sizes.

## 6 Discussion

Overall performance at the largest training sizes is generally good for all neural network models in the study, though the systems diverge substantially at lower training sizes, where the character transformer CLUZH greatly outperforms the other models on all splits on all languages. Evaluation across several random seeds highlights the relative low sensitivity of the systems to sampling effects, and this is likely cognitively plausible given the uniformity of language acquisition. However, the choice of sampling strategy had a more meaningful effect, since UNIFORM sampling consistently inflated scores relative to WEIGHTED sampling. This is particularly apparent for English past tense learning, where the sampling strategy dramatically alters the number of irregular forms in the training data.

Our linguistic analysis focused on how well ANNs’ learning trajectories and error types correspond to observed human learning. Results on German were generally promising. Productive inflection patterns emerge in roughly the expected order, and CLUZH-B4 demonstrates over-application of *-s*. Results for Arabic and English were less promising, as models did not show patterns of *u*-shaped learning which are robustly attested in child learners, nor do they reproduce attested strong asymmetries between different error types.

Taken together, this work demonstrates that the field’s conclusions about ANNs as cognitive models of morphology are sensitive to the often implicit task design decisions made during evaluation. We hope that this line of work will be extended to more models, languages, and morphological phenomena.

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## A Accuracy Model

**Splittype** here refers to sampling strategy. **wueta1** refers to CHR-TRM, and **kced** refers to E-D.

Call:

```
glm(formula = totalAcc ~ trainsize * model * language * splitttype,
    family = quasibinomial(), data = data_select)
```

Deviance Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -0.80194 | -0.06361 | 0.00755 | 0.08182 | 0.41850 |

Coefficients:

|                         | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------------------|------------|------------|---------|--------------|
| (Intercept)             | -1.707e-01 | 1.015e-01  | -1.682  | 0.092953 .   |
| trainsize               | 9.702e-04  | 1.672e-04  | 5.803   | 8.74e-09 *** |
| modelcluzh-gr           | -4.679e-02 | 1.434e-01  | -0.326  | 0.744331     |
| modelkced               | -1.566e+00 | 1.559e-01  | -10.043 | < 2e-16 ***  |
| modelwueta1             | -1.058e+00 | 1.486e-01  | -7.116  | 2.12e-12 *** |
| languageDeN             | 9.819e-01  | 1.849e-01  | 5.311   | 1.35e-07 *** |
| languageEnV             | 1.581e+00  | 1.694e-01  | 9.331   | < 2e-16 ***  |
| splittypenaive_weighted | 3.516e-02  | 1.435e-01  | 0.245   | 0.806488     |



|                                                          |            |           |        |          |     |
|----------------------------------------------------------|------------|-----------|--------|----------|-----|
| trainsize:modelcluzh-gr                                  | 2.184e-06  | 2.360e-04 | 0.009  | 0.992619 |     |
| trainsize:modelkced                                      | 1.224e-03  | 2.467e-04 | 4.963  | 8.16e-07 | *** |
| trainsize:modelwueta1                                    | 1.167e-03  | 2.422e-04 | 4.819  | 1.67e-06 | *** |
| trainsize:languageN                                      | -8.857e-05 | 4.422e-04 | -0.200 | 0.841280 |     |
| trainsize:languageV                                      | -2.767e-04 | 2.862e-04 | -0.967 | 0.333983 |     |
| modelcluzh-gr:languageN                                  | 6.410e-02  | 2.616e-01 | 0.245  | 0.806483 |     |
| modelkced:languageN                                      | -6.797e-01 | 2.661e-01 | -2.554 | 0.010787 | *   |
| modelwueta1:languageN                                    | -1.285e+00 | 2.631e-01 | -4.885 | 1.21e-06 | *** |
| modelcluzh-gr:languageV                                  | 5.431e-02  | 2.395e-01 | 0.227  | 0.820681 |     |
| modelkced:languageV                                      | 1.295e-01  | 2.357e-01 | 0.550  | 0.582660 |     |
| modelwueta1:languageV                                    | -8.054e-01 | 2.298e-01 | -3.505 | 0.000478 | *** |
| trainsize:splitypenaive_weighted                         | -4.566e-05 | 2.364e-04 | -0.193 | 0.846851 |     |
| modelcluzh-gr:splitypenaive_weighted                     | -3.560e-02 | 2.029e-01 | -0.175 | 0.860736 |     |
| modelkced:splitypenaive_weighted                         | -3.501e-01 | 2.242e-01 | -1.562 | 0.118676 |     |
| modelwueta1:splitypenaive_weighted                       | -2.763e-01 | 2.120e-01 | -1.303 | 0.192791 |     |
| languageN:splitypenaive_weighted                         | -3.175e-01 | 2.589e-01 | -1.226 | 0.220415 |     |
| languageV:splitypenaive_weighted                         | -1.518e-01 | 2.388e-01 | -0.636 | 0.525069 |     |
| trainsize:modelcluzh-gr:languageN                        | -4.371e-05 | 6.252e-04 | -0.070 | 0.944281 |     |
| trainsize:modelkced:languageN                            | 2.540e-03  | 6.235e-04 | 4.073  | 5.01e-05 | *** |
| trainsize:modelwueta1:languageN                          | 3.016e-03  | 6.281e-04 | 4.802  | 1.81e-06 | *** |
| trainsize:modelcluzh-gr:languageV                        | -2.609e-05 | 4.042e-04 | -0.065 | 0.948546 |     |
| trainsize:modelkced:languageV                            | 6.946e-04  | 4.050e-04 | 1.715  | 0.086652 | .   |
| trainsize:modelwueta1:languageV                          | 1.165e-03  | 3.997e-04 | 2.915  | 0.003642 | **  |
| trainsize:modelcluzh-gr:splitypenaive_weighted           | 5.712e-05  | 3.338e-04 | 0.171  | 0.864140 |     |
| trainsize:modelkced:splitypenaive_weighted               | 4.544e-04  | 3.528e-04 | 1.288  | 0.198072 |     |
| trainsize:modelwueta1:splitypenaive_weighted             | 3.356e-04  | 3.446e-04 | 0.974  | 0.330401 |     |
| trainsize:languageN:splitypenaive_weighted               | 6.620e-04  | 6.230e-04 | 1.063  | 0.288210 |     |
| trainsize:languageV:splitypenaive_weighted               | 3.087e-04  | 4.068e-04 | 0.759  | 0.448050 |     |
| modelcluzh-gr:languageN:splitypenaive_weighted           | 2.287e-02  | 3.662e-01 | 0.062  | 0.950210 |     |
| modelkced:languageN:splitypenaive_weighted               | 5.563e-01  | 3.772e-01 | 1.475  | 0.140560 |     |
| modelwueta1:languageN:splitypenaive_weighted             | 2.504e-01  | 3.739e-01 | 0.670  | 0.503228 |     |
| modelcluzh-gr:languageV:splitypenaive_weighted           | 3.828e-02  | 3.376e-01 | 0.113  | 0.909755 |     |
| modelkced:languageV:splitypenaive_weighted               | -8.310e-01 | 3.364e-01 | -2.470 | 0.013681 | *   |
| modelwueta1:languageV:splitypenaive_weighted             | -7.806e-01 | 3.286e-01 | -2.376 | 0.017712 | *   |
| trainsize:modelcluzh-gr:languageN:splitypenaive_weighted | -4.187e-05 | 8.806e-04 | -0.048 | 0.962087 |     |
| trainsize:modelkced:languageN:splitypenaive_weighted     | -1.354e-03 | 8.784e-04 | -1.541 | 0.123608 |     |
| trainsize:modelwueta1:languageN:splitypenaive_weighted   | -7.161e-04 | 8.888e-04 | -0.806 | 0.420607 |     |
| trainsize:modelcluzh-gr:languageV:splitypenaive_weighted | -6.998e-05 | 5.743e-04 | -0.122 | 0.903045 |     |
| trainsize:modelkced:languageV:splitypenaive_weighted     | 9.120e-04  | 5.807e-04 | 1.571  | 0.116597 |     |
| trainsize:modelwueta1:languageV:splitypenaive_weighted   | 7.541e-04  | 5.723e-04 | 1.318  | 0.187857 |     |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 0.02697953)

Null deviance: 219.228 on 1039 degrees of freedom  
Residual deviance: 30.113 on 992 degrees of freedom  
AIC: NA

Number of Fisher Scoring iterations: 5

## B Score Range and Variability Models

### B.1 Score Range

Call:

```
lm(formula = range ~ trainsize * model * language * splittype,
    data = summary_clean)
```

Residuals:

| Min      | 1Q       | Median   | 3Q      | Max     |
|----------|----------|----------|---------|---------|
| -0.28736 | -0.06608 | -0.00874 | 0.01560 | 0.78366 |

Coefficients:

|                         | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------------------|------------|------------|---------|--------------|
| (Intercept)             | 3.095e-01  | 4.255e-02  | 7.275   | 7.30e-13 *** |
| trainsize               | -2.018e-04 | 6.857e-05  | -2.944  | 0.003322 **  |
| modelcluzh-gr           | 1.129e-02  | 6.017e-02  | 0.188   | 0.851228     |
| modelkced               | -2.835e-01 | 6.017e-02  | -4.712  | 2.82e-06 *** |
| modelwueta1             | -1.578e-01 | 6.017e-02  | -2.622  | 0.008874 **  |
| languageN               | -2.502e-01 | 8.278e-02  | -3.022  | 0.002578 **  |
| languageV               | -2.585e-01 | 6.727e-02  | -3.842  | 0.000130 *** |
| splitypenaive_weighted  | -2.908e-02 | 6.017e-02  | -0.483  | 0.628955     |
| trainsize:modelcluzh-gr | -1.310e-05 | 9.697e-05  | -0.135  | 0.892598     |
| trainsize:modelkced     | 2.584e-04  | 9.697e-05  | 2.665   | 0.007836 **  |
| trainsize:modelwueta1   | 1.866e-04  | 9.697e-05  | 1.924   | 0.054663 .   |
| trainsize:languageN     | 1.561e-04  | 1.948e-04  | 0.801   | 0.423047     |
| trainsize:languageV     | 2.049e-04  | 1.084e-04  | 1.890   | 0.059057 .   |

```

modelcluzh-gr:languageN -3.251e-02 1.171e-01 -0.278 0.781304
modelkced:languageN 3.431e-01 1.171e-01 2.931 0.003465 **
modelwuetal:languageN 2.002e-01 1.171e-01 1.710 0.087535 .
modelcluzh-gr:languageV -9.242e-03 9.513e-02 -0.097 0.922630
modelkced:languageV 3.238e-01 9.513e-02 3.404 0.000692 ***
modelwuetal:languageV 1.856e-01 9.513e-02 1.951 0.051388 .
trainsize:splittpenaive_weighted 6.429e-05 9.697e-05 0.663 0.507519
modelcluzh-gr:splittpenaive_weighted 1.271e-02 8.509e-02 0.149 0.881278
modelkced:splittpenaive_weighted 3.466e-02 8.509e-02 0.407 0.683892
modelwuetal:splittpenaive_weighted 6.085e-02 8.509e-02 0.715 0.474746
languageN:splittpenaive_weighted 1.644e-01 1.171e-01 1.404 0.160517
languageV:splittpenaive_weighted 4.160e-02 9.513e-02 0.437 0.662028
trainsize:modelcluzh-gr:languageN 6.024e-05 2.755e-04 0.219 0.826963
trainsize:modelkced:languageN -3.579e-04 2.755e-04 -1.299 0.194188
trainsize:modelwuetal:languageN -2.594e-04 2.755e-04 -0.942 0.346608
trainsize:modelcluzh-gr:languageV 1.200e-05 1.533e-04 0.078 0.937621
trainsize:modelkced:languageV -3.091e-04 1.533e-04 -2.016 0.044115 *
trainsize:modelwuetal:languageV -2.181e-04 1.533e-04 -1.422 0.155270
trainsize:modelcluzh-gr:splittpenaive_weighted -1.989e-05 1.371e-04 -0.145 0.884729
trainsize:modelkced:splittpenaive_weighted 1.347e-06 1.371e-04 0.010 0.992167
trainsize:modelwuetal:splittpenaive_weighted 6.053e-05 1.371e-04 0.441 0.659029
trainsize:languageN:splittpenaive_weighted -3.462e-04 2.755e-04 -1.257 0.209201
trainsize:languageV:splittpenaive_weighted -1.018e-04 1.533e-04 -0.664 0.506836
modelcluzh-gr:languageN:splittpenaive_weighted -5.267e-03 1.656e-01 -0.032 0.974627
modelkced:languageN:splittpenaive_weighted -3.910e-02 1.656e-01 -0.236 0.813351
modelwuetal:languageN:splittpenaive_weighted -2.055e-01 1.656e-01 -1.241 0.214799
modelcluzh-gr:languageV:splittpenaive_weighted -1.813e-02 1.345e-01 -0.135 0.892853
modelkced:languageV:splittpenaive_weighted -2.744e-02 1.345e-01 -0.204 0.838458
modelwuetal:languageV:splittpenaive_weighted -8.912e-02 1.345e-01 -0.662 0.507852
trainsize:modelcluzh-gr:languageN:splittpenaive_weighted 8.935e-06 3.896e-04 0.023 0.981709
trainsize:modelkced:languageN:splittpenaive_weighted 8.177e-06 3.896e-04 0.021 0.983260
trainsize:modelwuetal:languageN:splittpenaive_weighted 2.647e-04 3.896e-04 0.679 0.497038
trainsize:modelcluzh-gr:languageV:splittpenaive_weighted 2.317e-05 2.168e-04 0.107 0.914929
trainsize:modelkced:languageV:splittpenaive_weighted -9.880e-06 2.168e-04 -0.046 0.963668
trainsize:modelwuetal:languageV:splittpenaive_weighted -1.240e-05 2.168e-04 -0.057 0.954422
---

```

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1526 on 944 degrees of freedom

Multiple R-squared: 0.2151, Adjusted R-squared: 0.176

F-statistic: 5.505 on 47 and 944 DF, p-value: < 2.2e-16

## B.2 Score Variability

Call:

```
lm(formula = std ~ trainsize * model * language * splittype,
    data = summary_clean)
```

Residuals:

```

      Min       1Q   Median       3Q      Max
-0.16175 -0.03011 -0.00324  0.00629  0.46887

```

Coefficients:

```

(Intercept) 1.349e-01 2.176e-02 6.201 8.36e-10 ***
trainsize -9.434e-05 3.507e-05 -2.690 0.00727 **
modelcluzh-gr 5.184e-03 3.077e-02 0.168 0.86625
modelkced -1.238e-01 3.077e-02 -4.023 6.22e-05 ***
modelwuetal -7.032e-02 3.077e-02 -2.285 0.02252 *
languageN -1.105e-01 4.234e-02 -2.610 0.00919 **
languageV -1.134e-01 3.441e-02 -3.295 0.00102 **
splittpenaive_weighted 2.532e-02 3.077e-02 0.823 0.41086
trainsize:modelcluzh-gr -6.924e-06 4.960e-05 -0.140 0.88900
trainsize:modelkced 1.175e-04 4.960e-05 2.370 0.01800 *
trainsize:modelwuetal 8.499e-05 4.960e-05 1.714 0.08692 .
trainsize:languageN 7.407e-05 9.963e-05 0.743 0.45742
trainsize:languageV 9.288e-05 5.545e-05 1.675 0.09426 .
modelcluzh-gr:languageN -1.154e-02 5.987e-02 -0.193 0.84722
modelkced:languageN 1.493e-01 5.987e-02 2.494 0.01279 *
modelwuetal:languageN 8.143e-02 5.987e-02 1.360 0.17414
modelcluzh-gr:languageV -4.914e-03 4.866e-02 -0.101 0.91957
modelkced:languageV 1.375e-01 4.866e-02 2.827 0.00480 **
modelwuetal:languageV 7.949e-02 4.866e-02 1.634 0.10264
trainsize:splittpenaive_weighted -8.089e-06 4.960e-05 -0.163 0.87047
modelcluzh-gr:splittpenaive_weighted 8.538e-03 4.352e-02 0.196 0.84450
modelkced:splittpenaive_weighted -2.184e-02 4.352e-02 -0.502 0.61589
modelwuetal:splittpenaive_weighted 1.623e-02 4.352e-02 0.373 0.70928
languageN:splittpenaive_weighted 2.897e-02 5.987e-02 0.484 0.62856
languageV:splittpenaive_weighted -2.098e-02 4.866e-02 -0.431 0.66643
trainsize:modelcluzh-gr:languageN 2.174e-05 1.409e-04 0.154 0.87743

```

|                                                          |            |           |        |         |
|----------------------------------------------------------|------------|-----------|--------|---------|
| trainsize:modelkced:languageN                            | -1.592e-04 | 1.409e-04 | -1.130 | 0.25880 |
| trainsize:modelwueta:languageN                           | -1.018e-04 | 1.409e-04 | -0.722 | 0.47024 |
| trainsize:modelcluzh-gr:languageV                        | 7.071e-06  | 7.842e-05 | 0.090  | 0.92817 |
| trainsize:modelkced:languageV                            | -1.328e-04 | 7.842e-05 | -1.693 | 0.09074 |
| trainsize:modelwueta:languageV                           | -9.444e-05 | 7.842e-05 | -1.204 | 0.22874 |
| trainsize:modelcluzh-gr:splitypenaive_weighted           | -1.295e-05 | 7.014e-05 | -0.185 | 0.85357 |
| trainsize:modelkced:splitypenaive_weighted               | 3.607e-05  | 7.014e-05 | 0.514  | 0.60716 |
| trainsize:modelwueta:splitypenaive_weighted              | 2.516e-05  | 7.014e-05 | 0.359  | 0.71993 |
| trainsize:languageN:splitypenaive_weighted               | -1.042e-04 | 1.409e-04 | -0.739 | 0.45993 |
| trainsize:languageV:splitypenaive_weighted               | -4.565e-06 | 7.842e-05 | -0.058 | 0.95359 |
| modelcluzh-gr:languageN:splitypenaive_weighted           | -7.423e-03 | 8.468e-02 | -0.088 | 0.93016 |
| modelkced:languageN:splitypenaive_weighted               | 2.039e-02  | 8.468e-02 | 0.241  | 0.80973 |
| modelwueta:languageN:splitypenaive_weighted              | -6.941e-02 | 8.468e-02 | -0.820 | 0.41258 |
| modelcluzh-gr:languageV:splitypenaive_weighted           | -9.160e-03 | 6.881e-02 | -0.133 | 0.89413 |
| modelkced:languageV:splitypenaive_weighted               | 2.637e-02  | 6.881e-02 | 0.383  | 0.70160 |
| modelwueta:languageV:splitypenaive_weighted              | -2.696e-02 | 6.881e-02 | -0.392 | 0.69532 |
| trainsize:modelcluzh-gr:languageN:splitypenaive_weighted | 1.150e-05  | 1.993e-04 | 0.058  | 0.95397 |
| trainsize:modelkced:languageN:splitypenaive_weighted     | -3.582e-05 | 1.993e-04 | -0.180 | 0.85736 |
| trainsize:modelwueta:languageN:splitypenaive_weighted    | 9.898e-05  | 1.993e-04 | 0.497  | 0.61949 |
| trainsize:modelcluzh-gr:languageV:splitypenaive_weighted | 1.276e-05  | 1.109e-04 | 0.115  | 0.90840 |
| trainsize:modelkced:languageV:splitypenaive_weighted     | -4.320e-05 | 1.109e-04 | -0.390 | 0.69694 |
| trainsize:modelwueta:languageV:splitypenaive_weighted    | -8.177e-06 | 1.109e-04 | -0.074 | 0.94124 |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07802 on 944 degrees of freedom

Multiple R-squared: 0.1998, Adjusted R-squared: 0.1599

F-statistic: 5.014 on 47 and 944 DF, p-value: < 2.2e-16